



Data Science Is Not AI, but It Is Its Genesis, and Its New Frontier

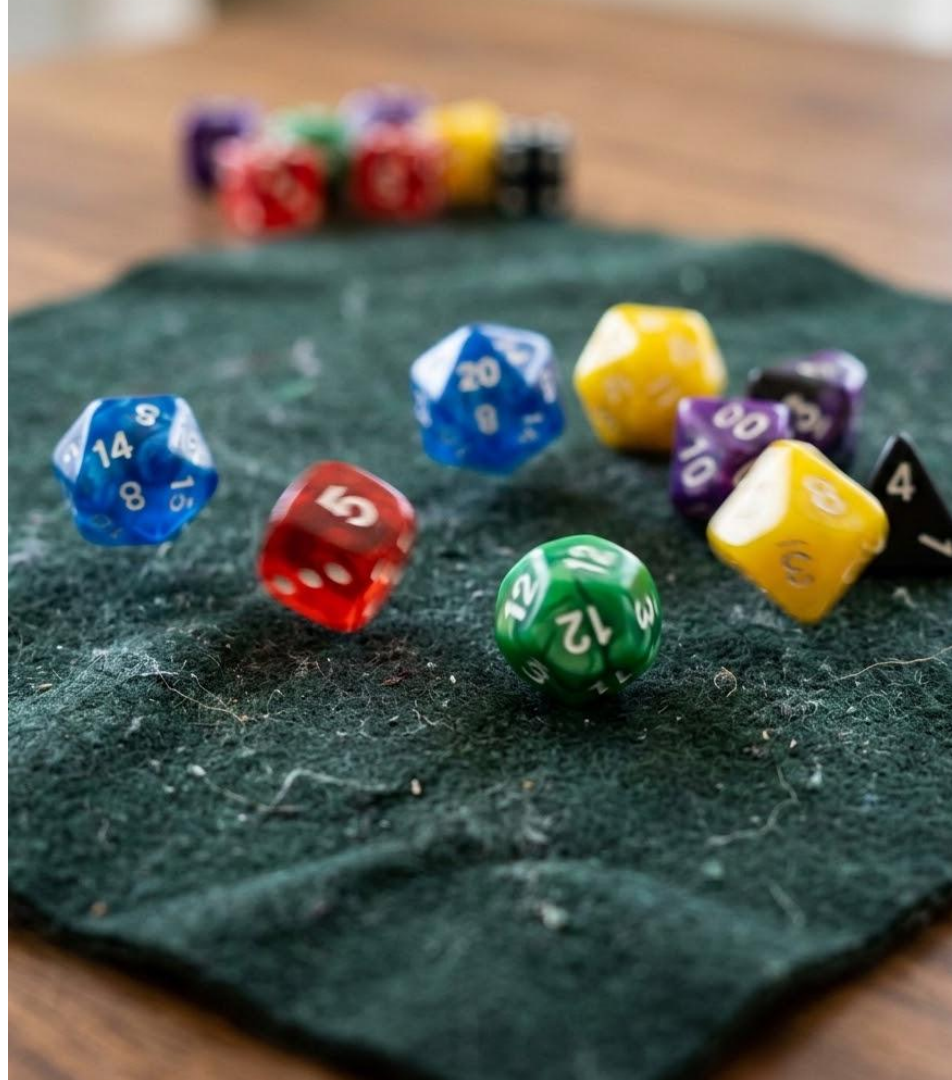
Gaël Varoquaux

CSO of Probabl, research director Inria

GOSIM Paris 2026

This talk

Statistical thinking for day-to-day AI



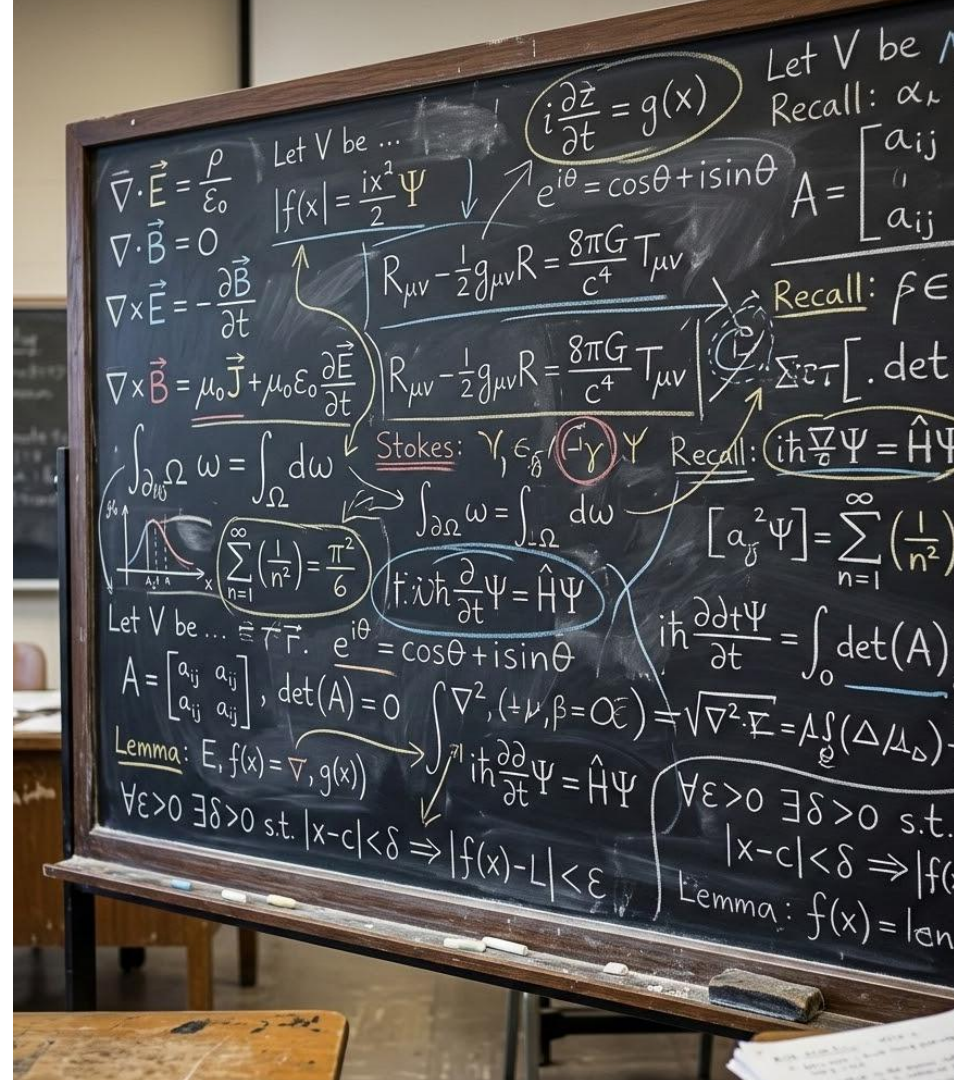


The statistics in AI

generate images, or text,
understand voice,
prove theorems

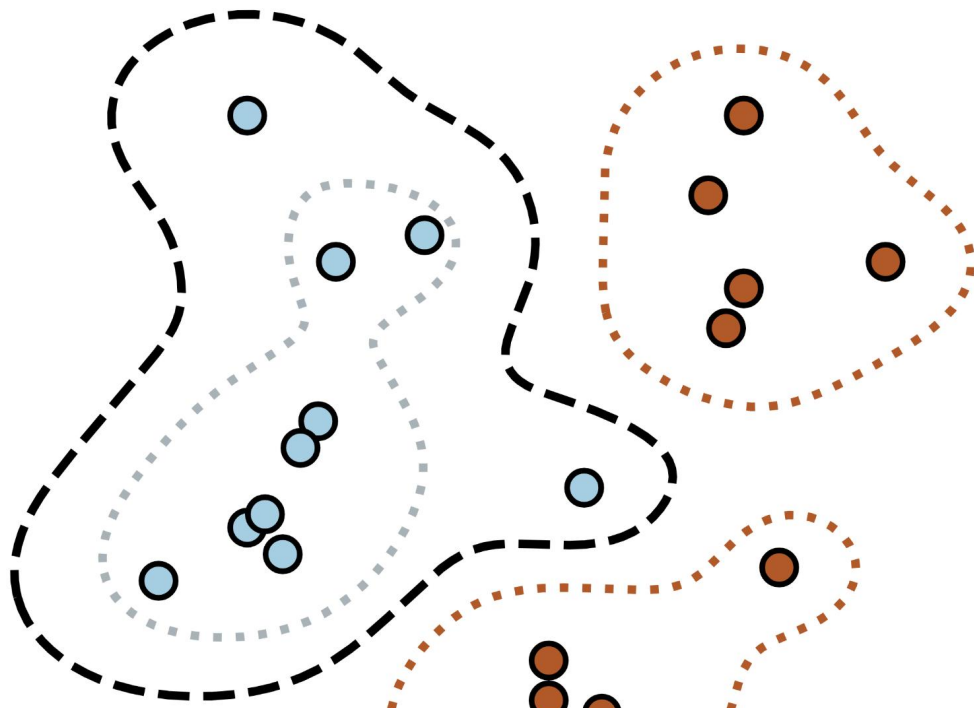
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Why statistical tools for deterministic problems?



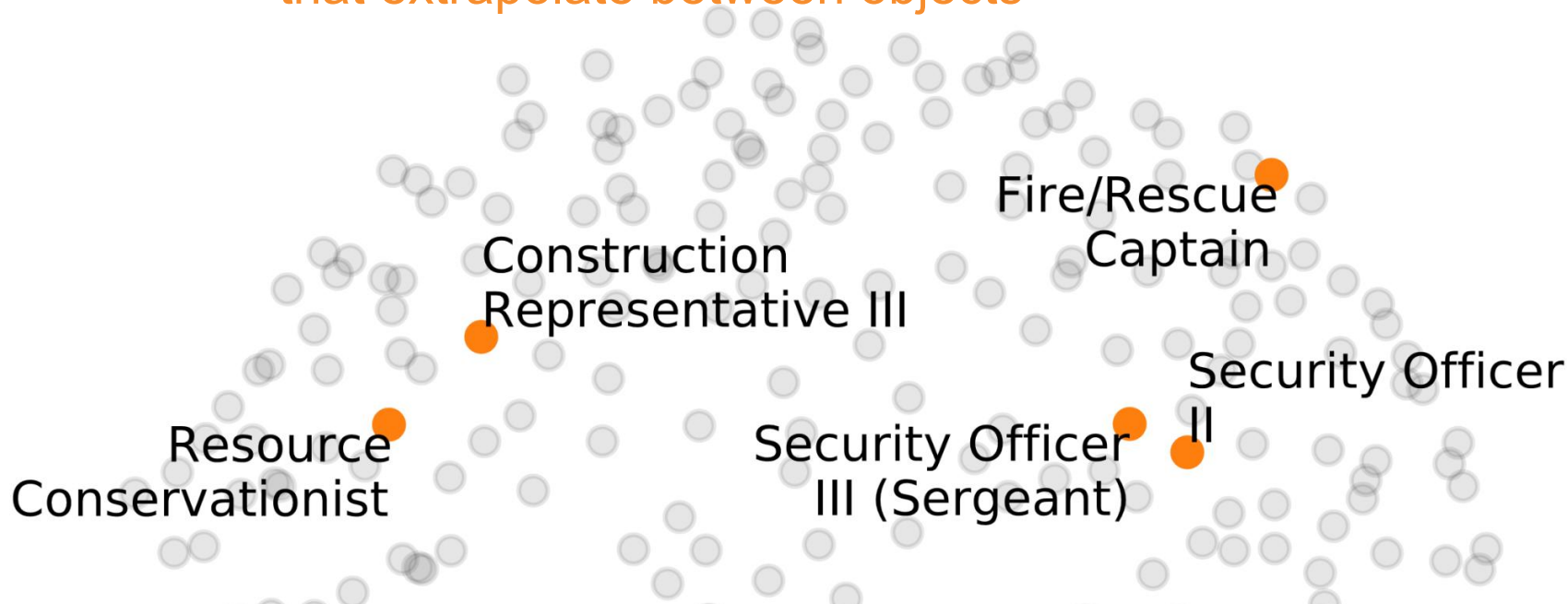
Finding continuities

Interpolation and extrapolation. AI leverages similarities.



Embedding discrete objects, symbols, logic

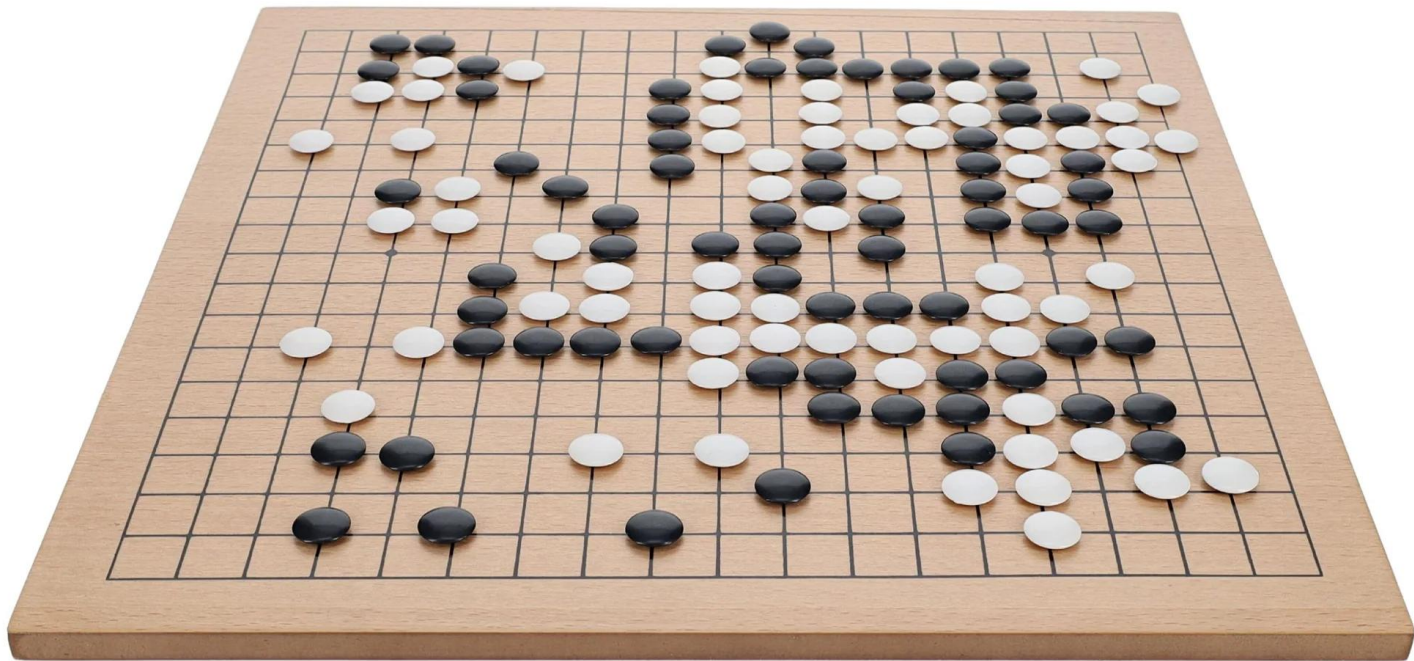
It endows them with a metric
which enables building predictive functions
that extrapolate between objects



Uncertainty from incomplete problem description

Ignoring some known details helps generalizing

We might not know everything about the situation



Ambiguities

Real life is full of them



Meaning beyond symbols

Context is key

A river *bank*

Bank of America

Disambiguation will only go so far

Fruit flies like a banana. Time flies like an arrow.



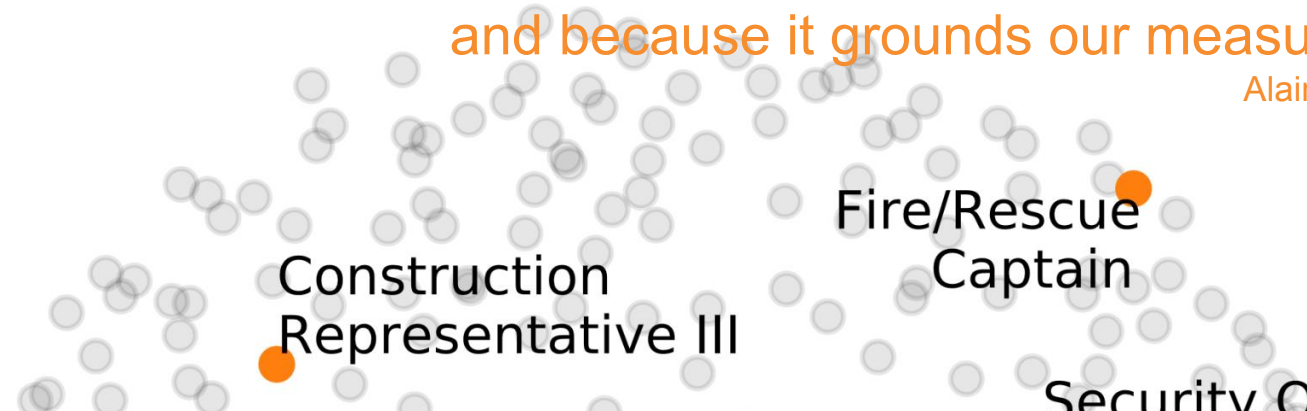
I don't believe logic, curation, normalization, rules
suffice to model the world

The arbitrary in categories

Junior Professor, Adjunct Professor, Assistant Professor, Tenured Track,
Senior Lecturer, Reader, Maître de conférence, Chargé de Recherche,
HDR, Universitätsprofessur, Assistenzprofessur, S-Professur,
Científico Titular, Profesor Ayudante Doctor, Profesor Colaborador,
Profesor Agregado, 兼职教授

But a system of categories ends up shaping the world
because the world needs to conform to it
and because it grounds our measurements

Alain Desrosières



Construction
Representative III

Fire/Rescue
Captain

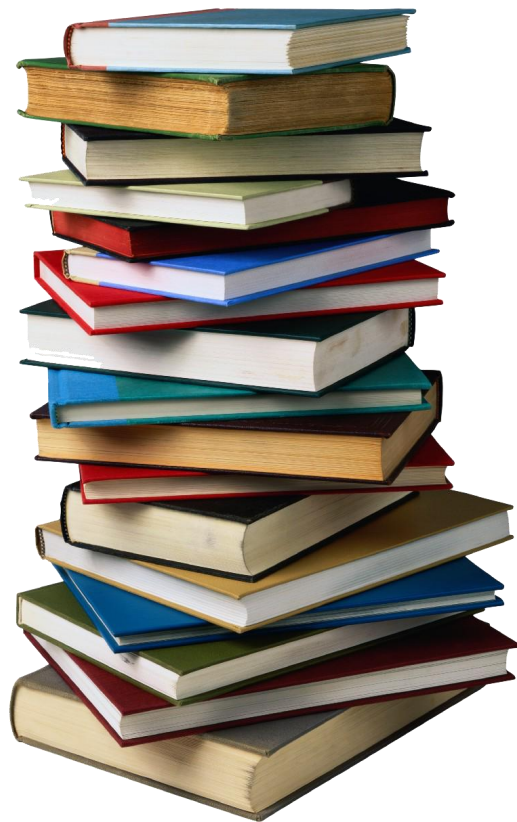
Security Officer

Compiling and compressing knowledge

A large language model builds a consistent view across

- misaligned references and schemas
- contradictory information
- non-quantified cultural uncertainties

Its limited memory forces lossy compression
thanks to continuous embeddings



Errors in AI outputs

Tell me something about Albert Einstein e.g., a short bio with the birth date and place

Albert Einstein was a German-born theoretical physicist who is widely held to be one of the greatest scientists of all time. Born in the German Empire, on January 14, 1879, Einstein grew up in Ulm, Germany. In 1905, he submitted a successful PhD dissertation to the Humboldt University of Berlin

Errors in AI outputs

We'd like a form of confidence

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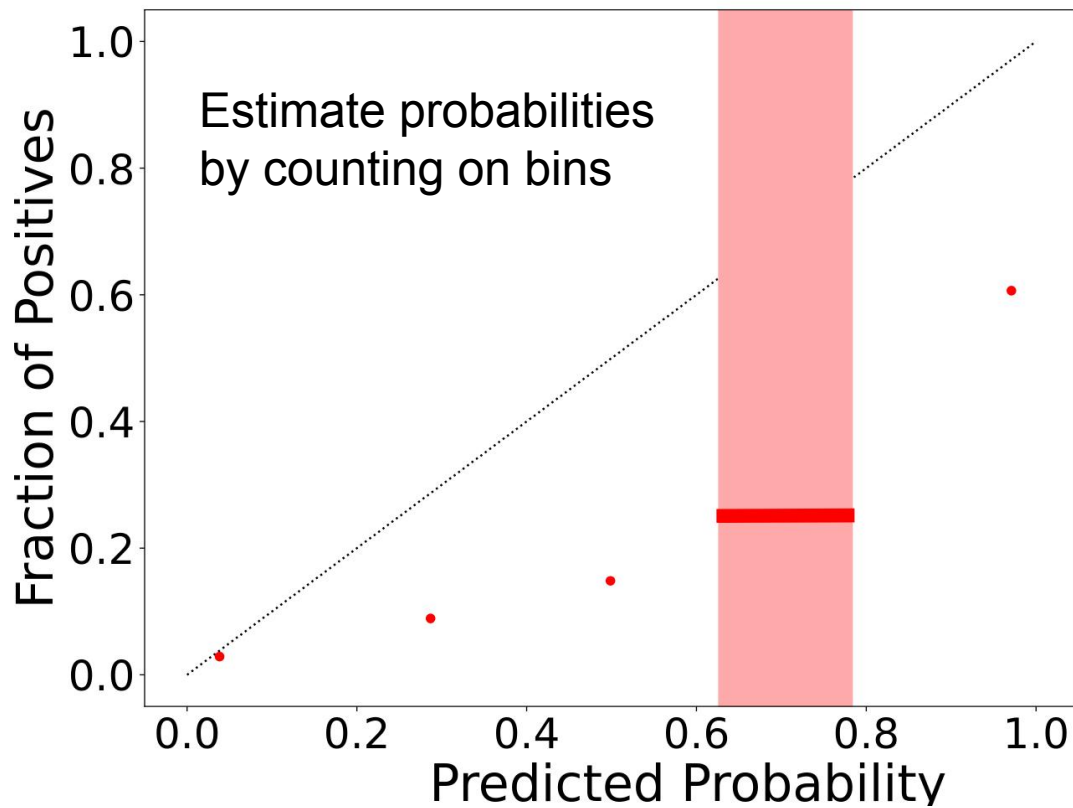
High confidence

Medium confidence

Low confidence

Errors in AI outputs

We'd like a form of confidence



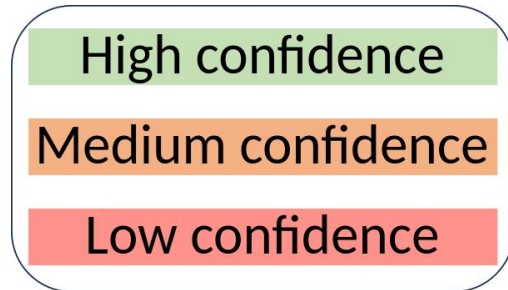
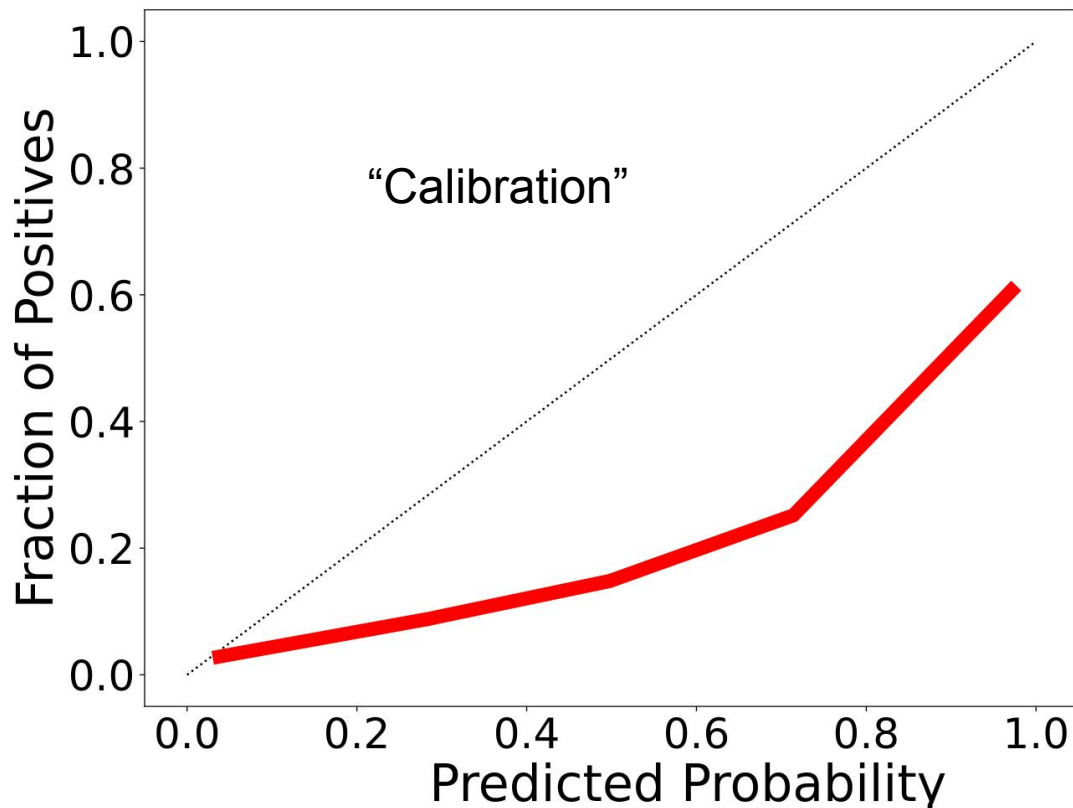
High confidence

Medium confidence

Low confidence

Errors in AI outputs

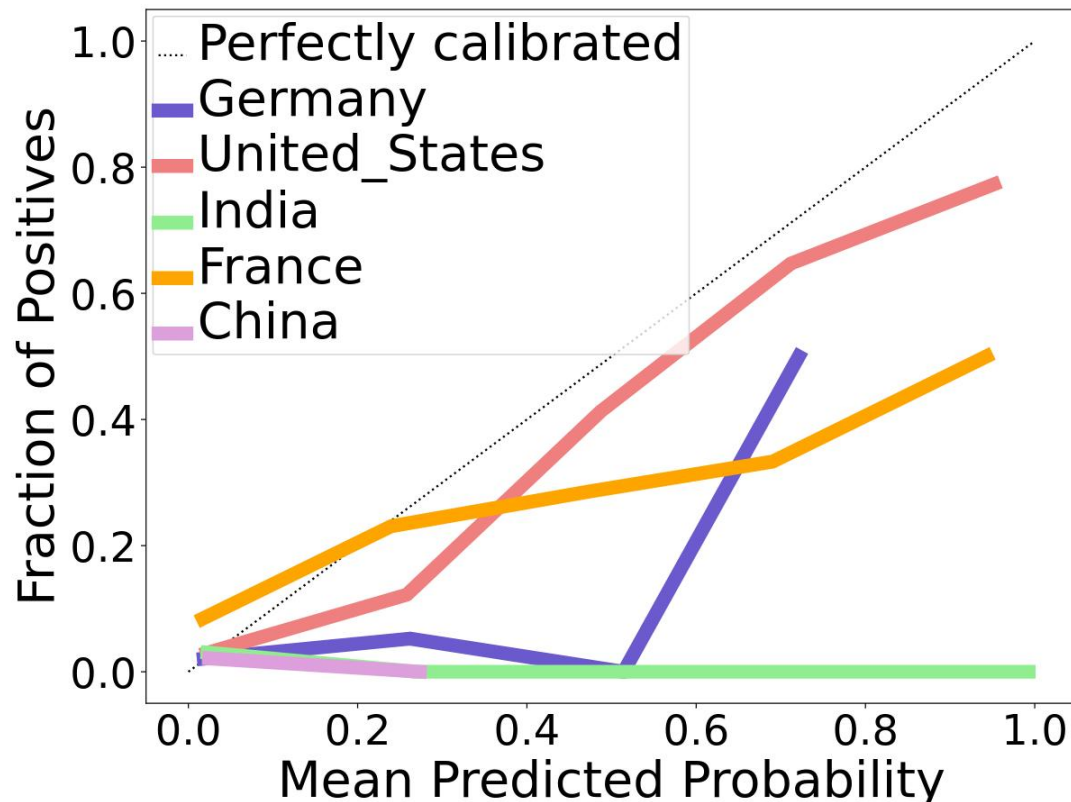
We'd like a form of confidence



Mistral 7B, questions on birth dates [Chen, Perez-Lebel, Suchanek, Varoquaux 2024]

Errors in AI outputs: Cultural structure

Statistics: Controlling the probability of the output given the input

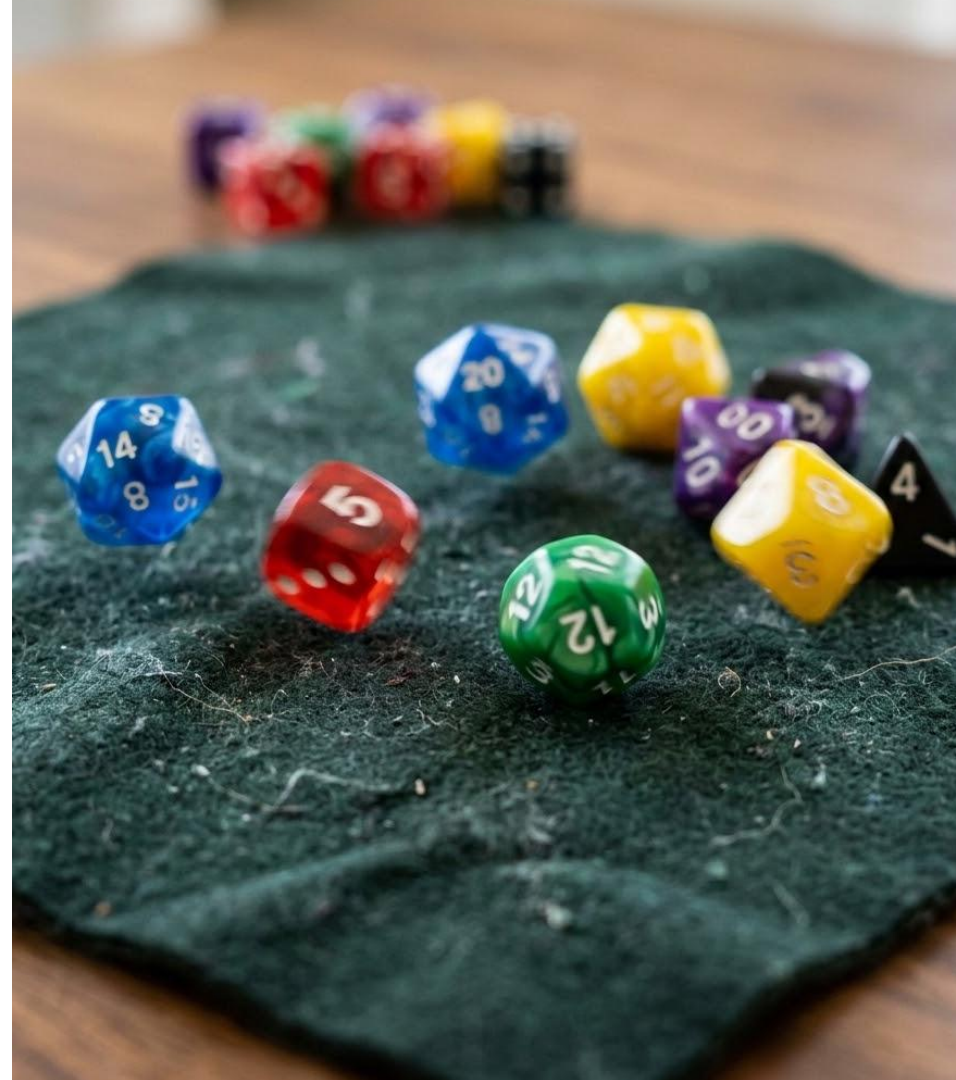


Varying the queried nationality

Mistral 7B, questions on birth dates [Chen,Perez-Lebel, Suchanek, Varoquaux 2024]

Statistics are
important for AI

Even for topics that come with
certainties





The science of data science

Data science

Build process from empirical evidence

Machine learning can automate
problems we don't know much
about



Many day-to-day applications often with high value

- Optimizing an industrial process
- Detecting credit card fraud
- Evaluating health risks
- Predicting airplane delays
- Detecting cyber attacks

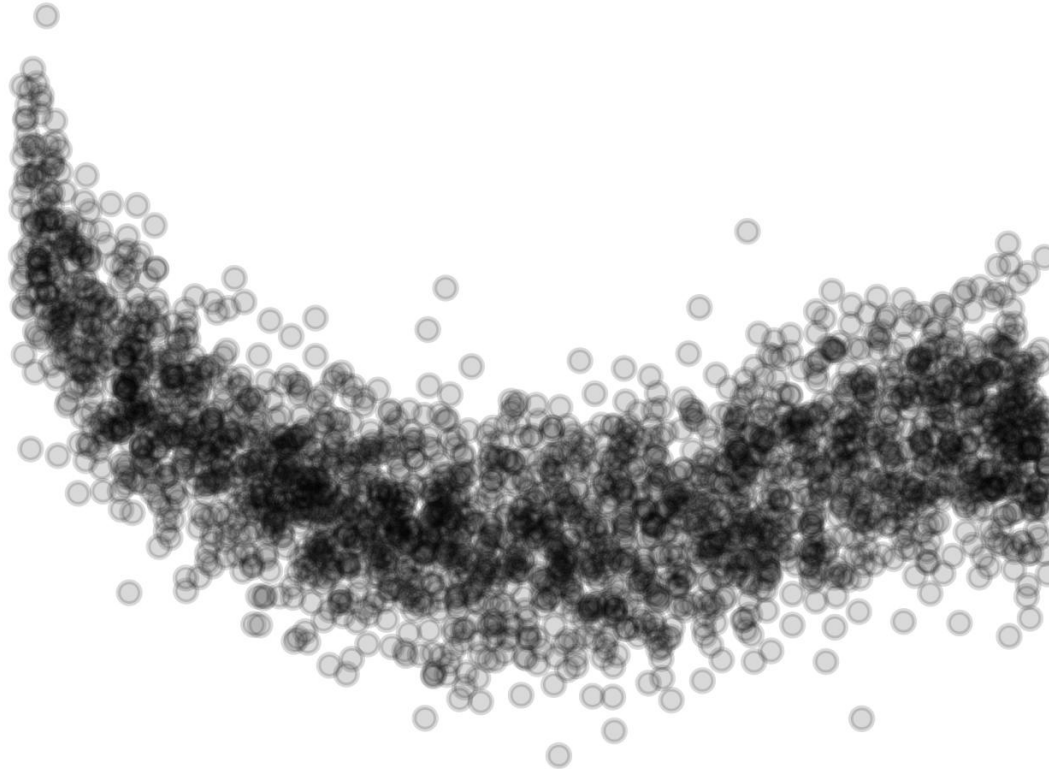
Tabular AI

Machine learning is central to this endeavor



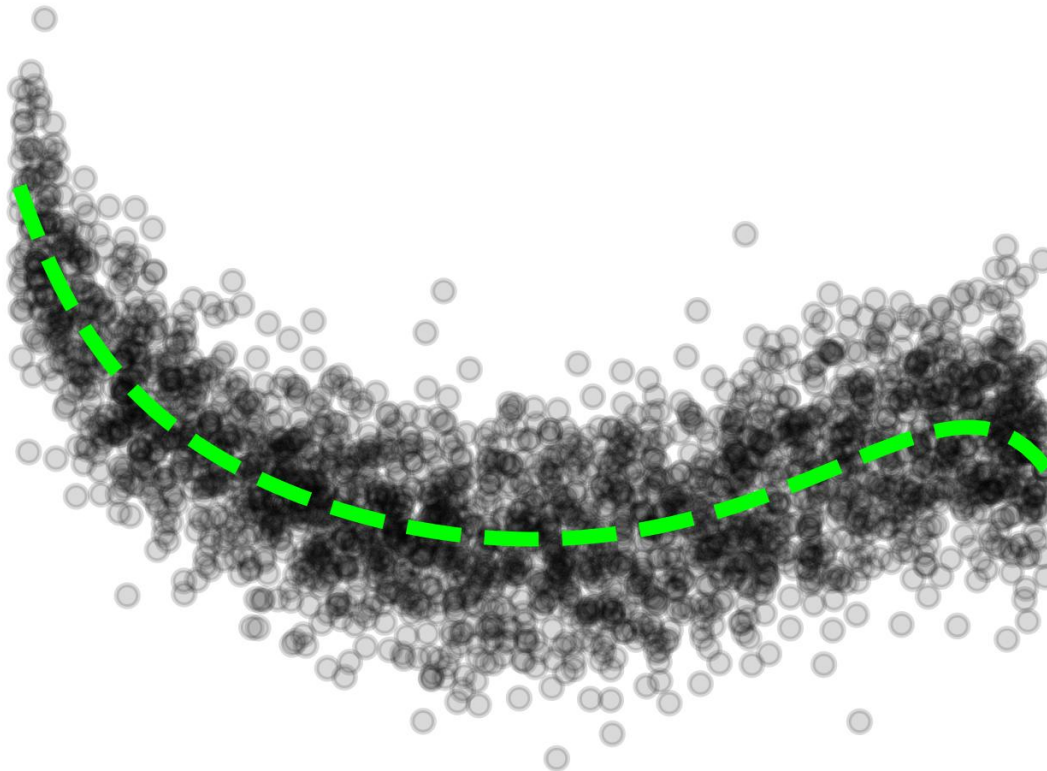
A statistical view on machine learning

- Given data (x, y) : input, output



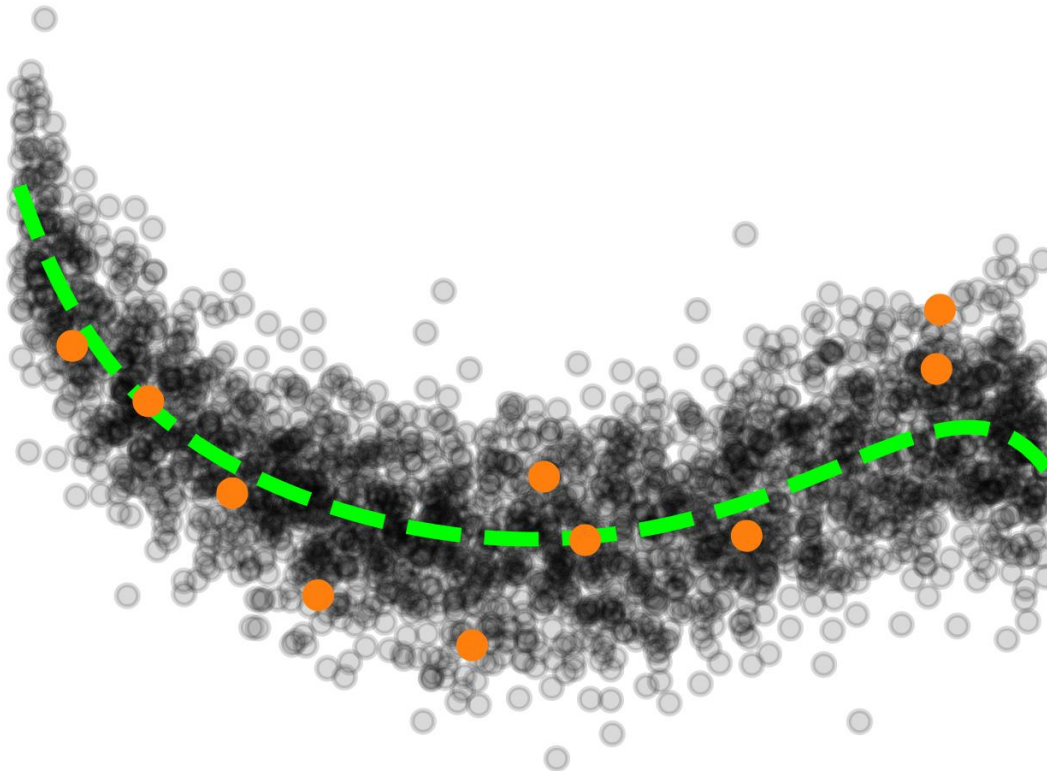
A statistical view on machine learning

- Given data (x, y) : input, output
- Our goal is to learn a function f :
 $x \rightarrow y$ such that $f(x) \approx y$
 - Typically: $f(x) = E[y|x]$
(quadratic loss)



A statistical view on machine learning

- Given data (x, y) : input, output
- Our goal is to learn a function f :
 $x \rightarrow y$ such that $f(x) \approx y$
 - Typically: $f(x) = E[y|x]$
(quadratic loss)
- Challenge is estimating the expectations



Aleatoric error: Not everything can be predicted

Modeling error versus intrinsic uncertainty

- **Epistemic error:** modeling imperfection
- **Aleatoric error:** intrinsic randomness across the data

Data science often has much aleatoric
Other domains of AI much less

But medical chatbots?

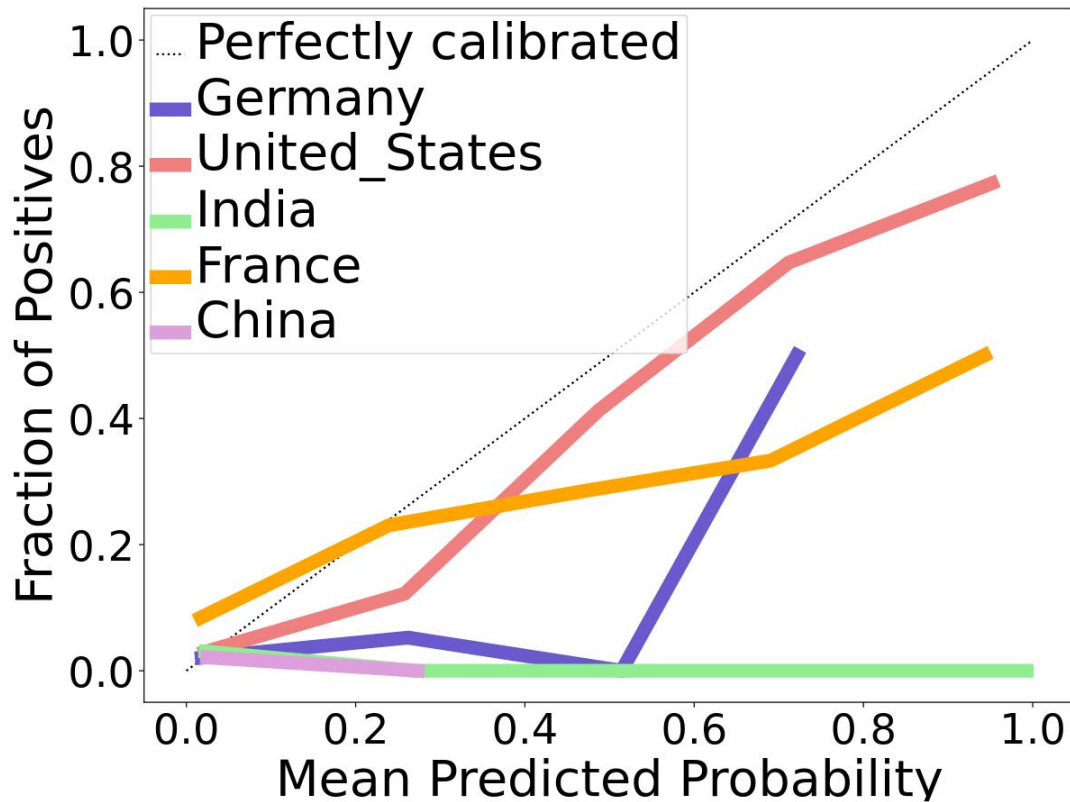


Accounting for aleatorics versus epistemic

Better decision: improve model, or data?

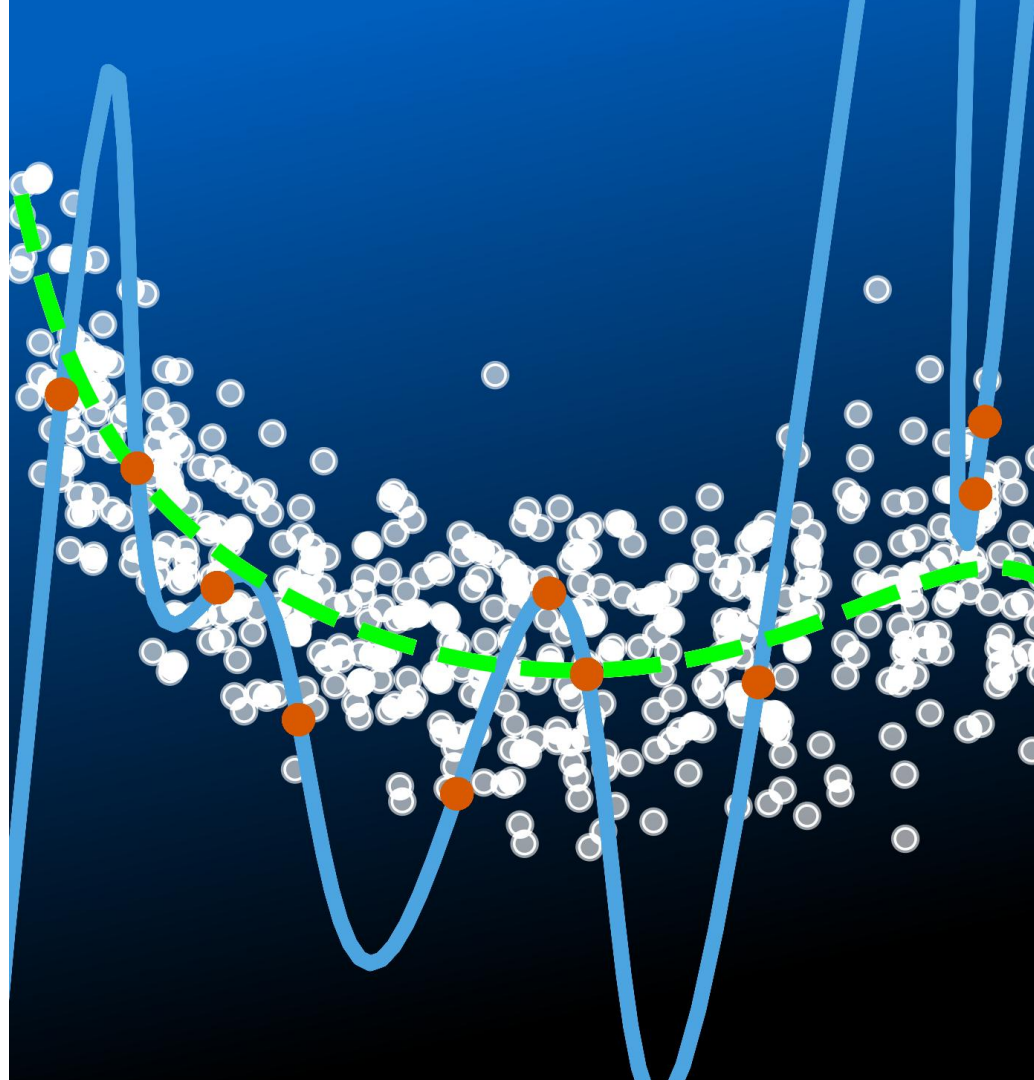
Predictable structure in model error
measures epistemic error

Work with [Melo, Perez-
Lebel, Le Morvan,
Varoquaux 2022, 2026]



With plenty of data,
optimal model?

Powerful models and big data
are all we need?

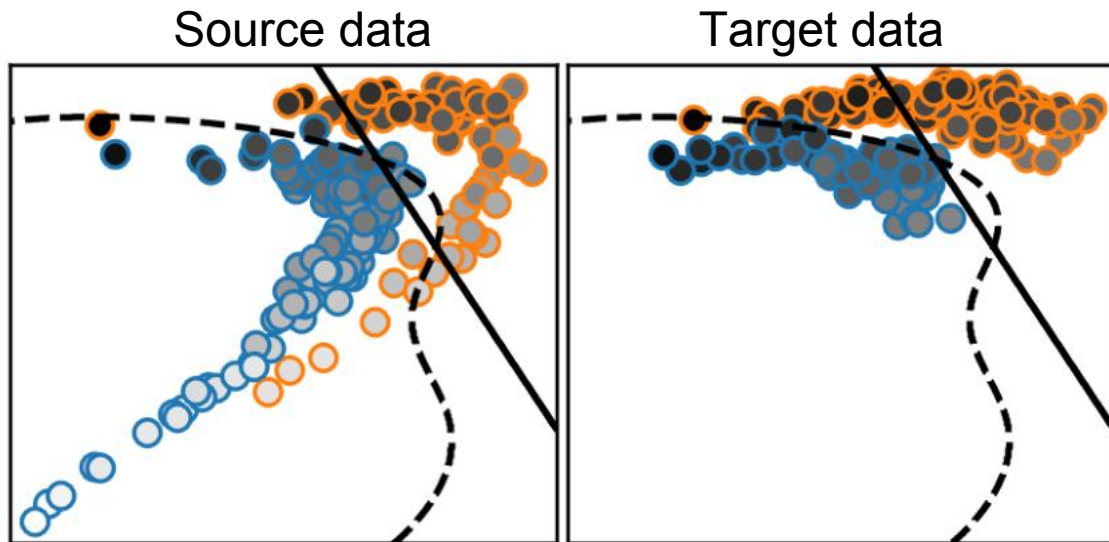


Distribution shifts

We seldom have the right data

Benign selection bias:
covariate shift

The covariate x changes,
but the link $x \rightarrow y$ is preserved



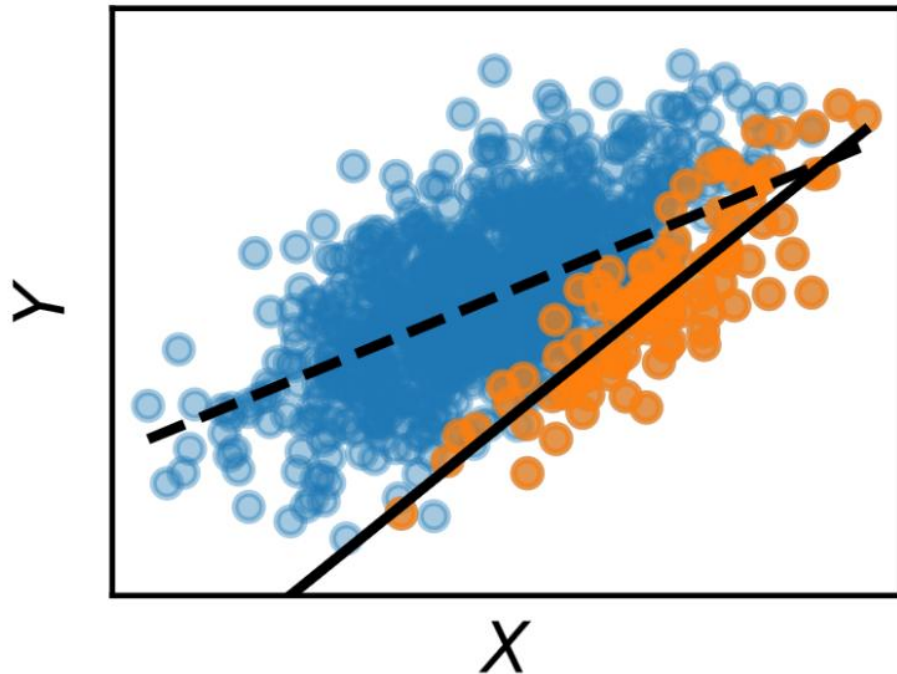
- If target domain is covered
- Flexible model and large data suffice

When selection bias breaks association

Data selection on a common mechanism of X and Y

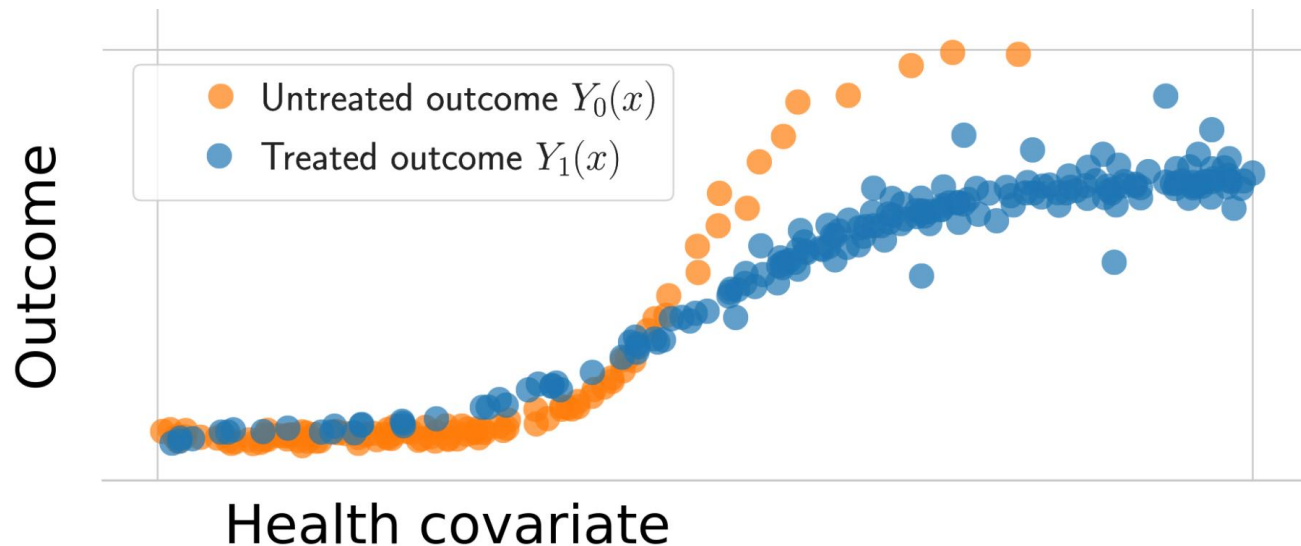
Distorts the association between X and Y

More data, bigger models won't solve the problem



Often prediction is for intervention

Causal Effects



Prediction for decision making:
Causal effects

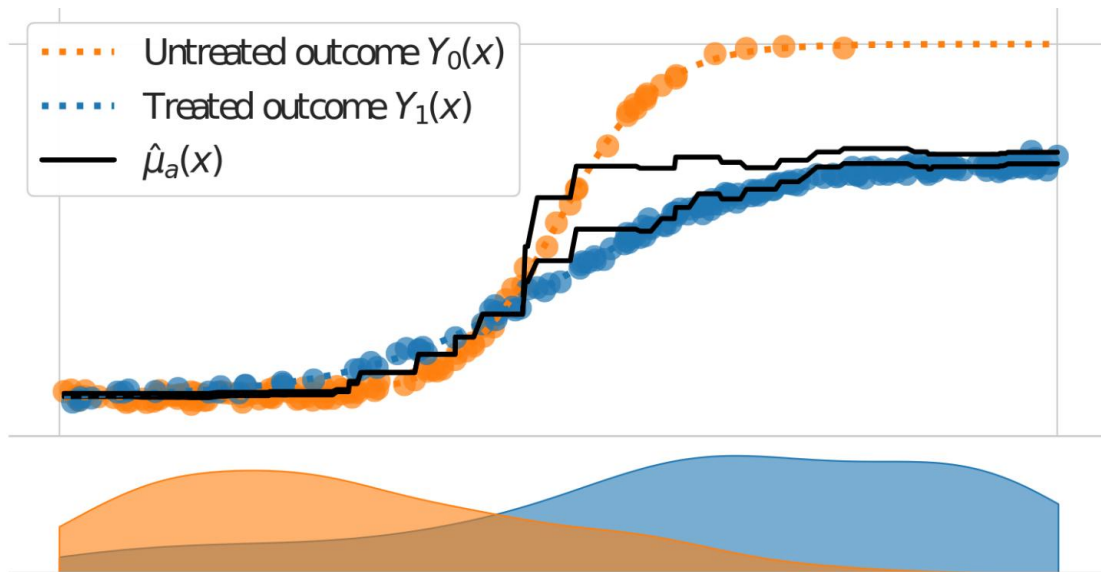
- Can a predictive model oriented intervention choices?
- We need the outcome as function of an intervention of interest

Often prediction is for intervention

Causal Effects

A selection bias:

- Healthy individuals did not receive the treatment
- Good risk minimizer associates treatment to negative health outcome

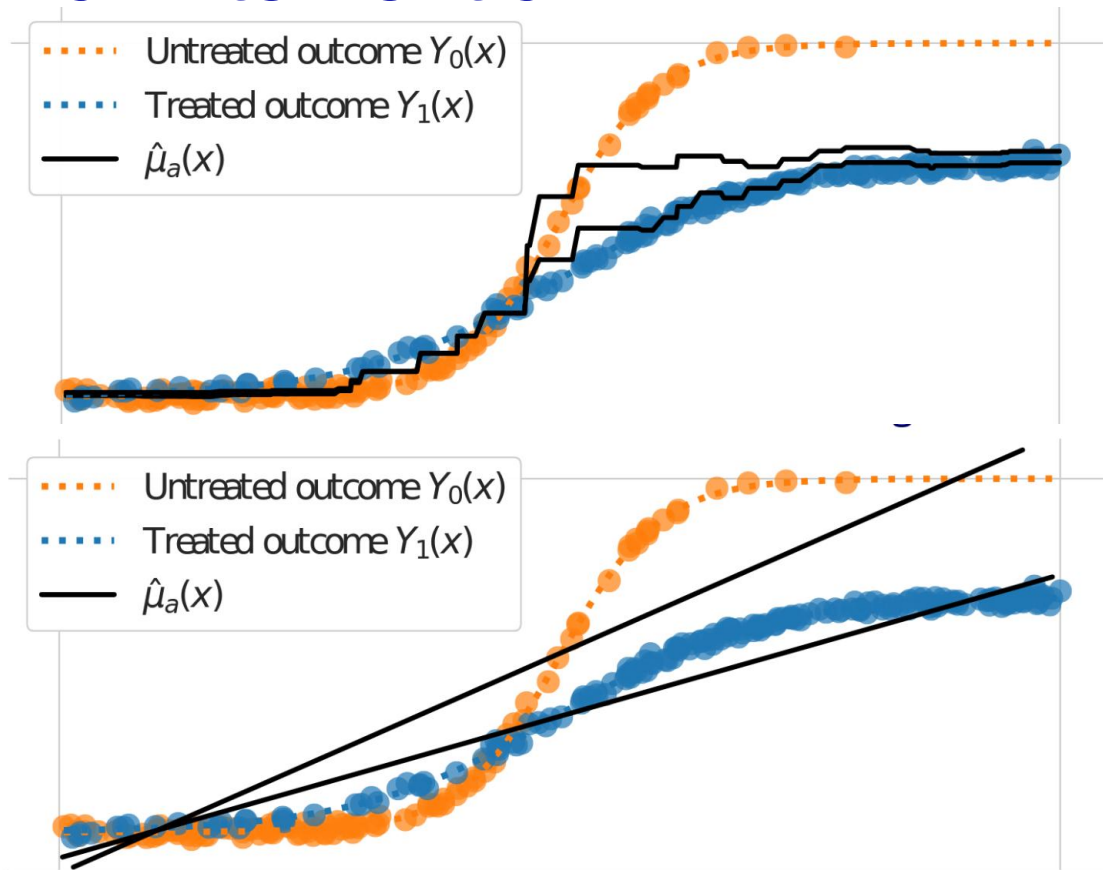


Often prediction is for intervention

Causal Effects

A selection bias:

- Healthy individuals did not receive the treatment
- Good risk minimizer associates treatment to negative health outcome
- The extrapolation is what matters



The science of data science

- Machine learning and prediction enables operating from empirics
- But data science is about much more than mere prediction from IID data
 - Selection biases (The data are never what we want)
 - Causal thinking





Validation is a bottleneck

The Golden Rule of AI Development

Anything goes, but we evaluate models on unseen data

In practice:

- Train/test split — the minimum
- Cross-validation — more robust estimates
- Temporal splits — for time-dependent data
- Out-of-distribution evaluation — stress testing
- Mimic the usage scenario

Common pitfall:

Data leakage — when test information sneaks into training,
often via preparation steps (prepare data with skrub)
performance metrics become misleading.

Which metric to optimize?

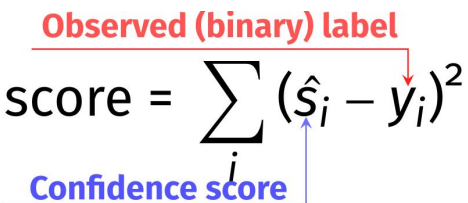
Accuracy

The simplest metric — but misleading with imbalanced data, poor statistical properties

Controlling probabilities

Proper scoring rules, such as Brier Score — probabilistic predictions

$$\text{Brier score} = \sum_i (\hat{s}_i - y_i)^2$$



Observed (binary) label

Confidence score

Business value or application utility

Reflect the actual decisions, cost, and value structure

Metrics are imperfect views on utility

Choosing the right metric is as important as choosing the right model

Learning from the industry trenches

An industry view on model evaluation

Offline model performance is just a health check

- Value-Performance Saturation
- Proxy Over-optimization
- Effect of model prediction changes the system

150 Successful Machine Learning Models: 6 Lessons Learned at Booking.com

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Skore

Addressing the validation bottleneck

A Python library streamlined machine learning evaluation, built on scikit-learn.

Evaluation made right:

- Cross-validation reports
- Automated metric diagnostics
- Distribution shift detection
- Model comparison dashboards



by the scikit-learn founders

```
import sklearn
import skore
import skrub

model = skrub.tabular_pipeline(sklearn.linear_model.Ridge())
report_ridge = skore.evaluate(model, df, y, splitter=5)
report_ridge
```

Report for Ridge			
Results	Diagnostic	Estimator	Data Help
Metric		mean	std
R ²		0.774567	0.015075
RMSE		13820.748326	957.334295
MAE		8867.060088	351.696693
MAPE		0.131006	0.002386
Fit time (s)		0.811745	0.019921
Predict time (s)		0.186262	0.019338

Skore

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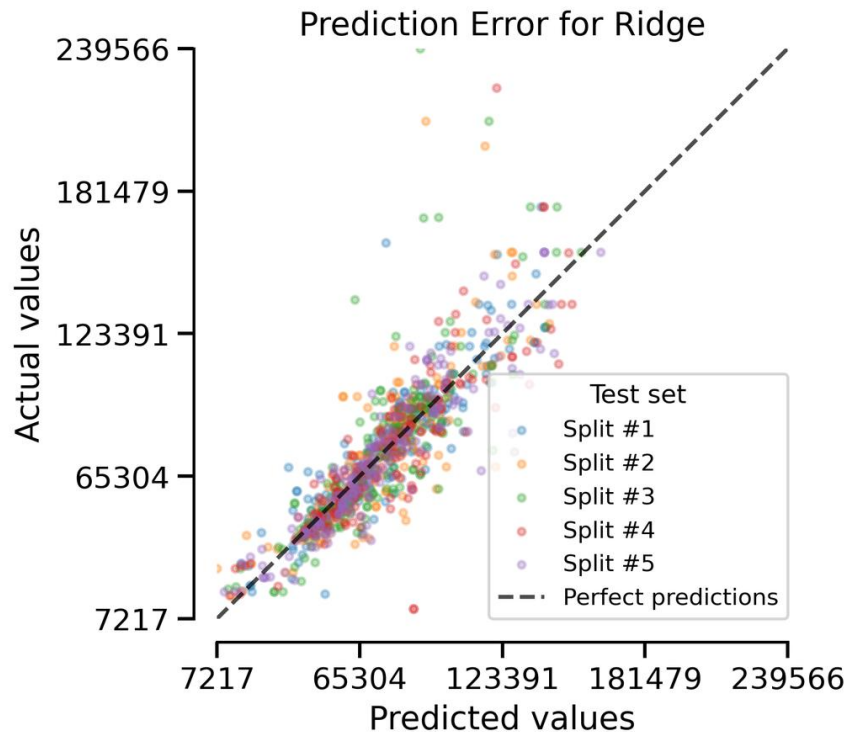
Evaluation made right:

- Cross-validation reports
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by the scikit-learn founders

```
error = report_ridge.metrics.prediction_error()  
error.plot(kind="actual_vs_predicted")
```



:probabl.

Skore

Addressing the validation bottleneck

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Evaluation made right:

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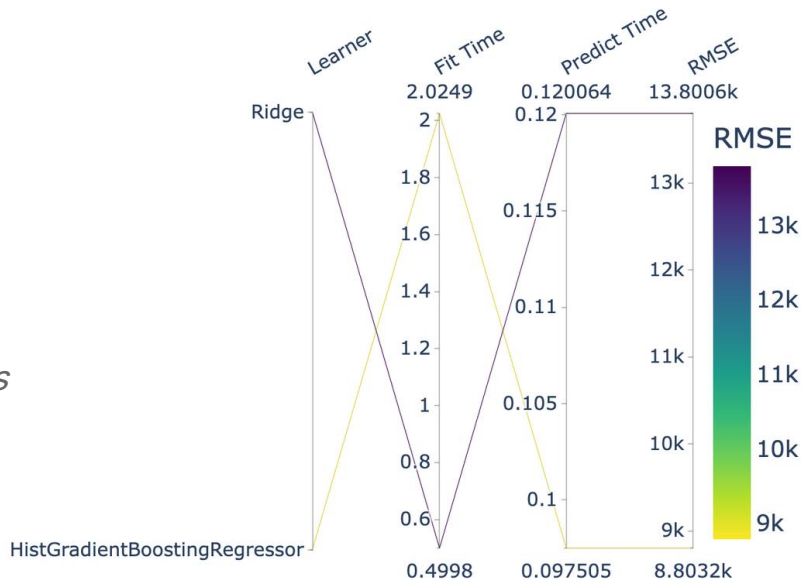


by the scikit-learn founders

```
project = skore.Project(  
    name="adult_census_survey", mode="local"  
)  
project.put("ridge", report_ridge)
```

Report Type: Task Type: Dataset:

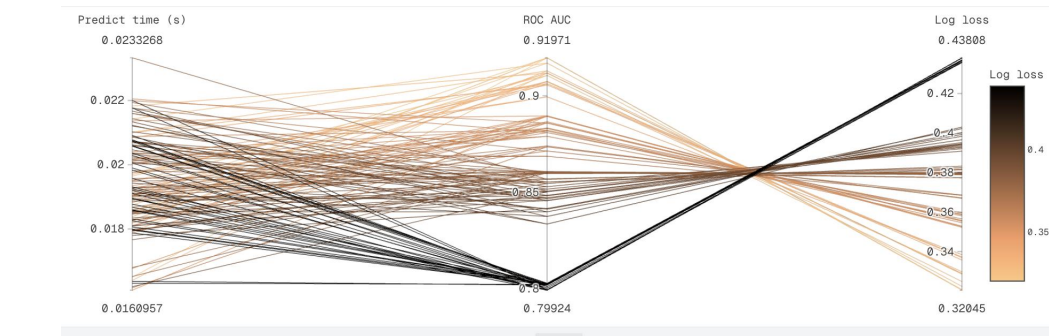
Color by:



:probabl.

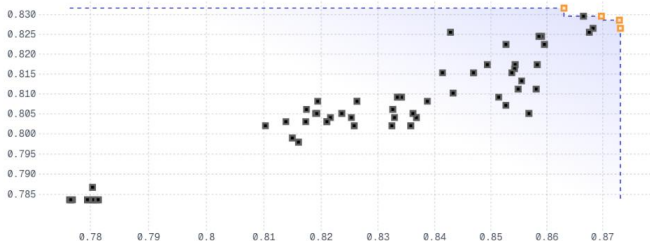
Skore: An online hub

To compare and inspect models and data



← Model finder

Pareto diagram



Selected trade-offs

RF (n=50, d=10, leaf=20)

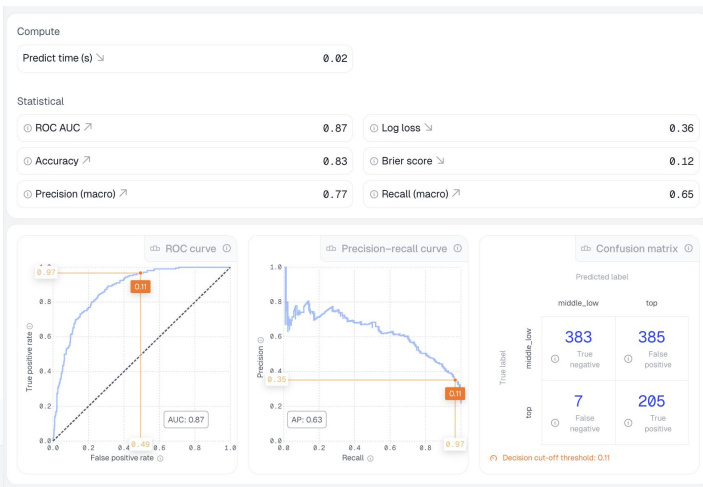
0.87 (ROC AUC) 0.83 (Accuracy)

No description.

RF (n=100, d=10, leaf=20)

0.87 ($-1.97e-4$) (ROC AUC) 0.83 ($+0.002$) (Accuracy)

No description.



Share with
other
stakeholders

:proba1.

AI is about automation
Data science is about custom data

Statistical thinking is needed

Minimizing train-test split error will only go so far
Application utility, Selection Biases, Causal thinking

Come along to the **Own Your Data Science & AI Workshop**
at the Open Stage.

:probabl.

